# NYMEX PLATINUM FUTURES (PL)

## Spread & Curve Structure -- Fundamental Reference

### South African Supply Concentration, Automotive Catalysis, and the Platinum-Palladium Substitution Relationship

This document presents a fact-based reference on the fundamental drivers of Platinum futures (ticker: PL), contract size 50 troy ounces, traded on NYMEX (a CME Group exchange) -- not COMEX, despite platinum sometimes being informally grouped with COMEX-traded precious metals in casual market commentary. Platinum is traded on the NYMEX division specifically, distinguishing it structurally from gold, silver, and copper (all traded on the COMEX division and covered in separate reference documents in this series), even though all are part of the same CME Group umbrella and trade on the same Globex electronic platform. Platinum is, among the precious metals covered in this document series, the one most heavily dependent on a single demand application (automotive catalytic converters) and the one with the most extreme supply-side geographic concentration (South Africa alone accounts for the large majority of global mine production), giving it a fundamental risk profile distinct from gold's monetary-asset framework and closer in some respects to an industrial metal with a single dominant, geographically concentrated, and politically exposed supply source. This document presents the platinum-specific fundamentals in full, with particular emphasis on South African mine supply and its labour and electricity-supply risk factors, the automotive catalyst demand structure, the platinum-palladium substitution relationship, and platinum's comparatively recent emergence as a hydrogen fuel cell catalyst material.

Futures First -- Internal Education Material
For Analyst Group Use

## 1. Contract Specifications and Exchange Clarification

Platinum futures trade on the NYMEX division of CME Group, a distinction worth establishing clearly given the common informal practice of referring to all CME Group precious and base metals collectively as "COMEX metals."

| Parameter | Detail |
|---|---|
| Exchange | NYMEX (New York Mercantile Exchange) -- a CME Group division. NOT COMEX: gold, silver, and copper trade on the COMEX division of CME Group, while platinum and palladium trade on the NYMEX division -- a real, if often blurred-in-casual-usage, distinction between the two divisions, both of which clear through the same CME Group clearing house and trade on the same Globex platform. |
| Ticker | PL (front month) |
| Contract size | 50 troy ounces |
| Quotation | US dollars per troy ounce (USD/oz); minimum tick = $0.10/oz = $5.00 per contract |
| Contract months | January, April, July, October as the primary actively quoted months (a quarterly cycle distinct from gold's more numerous monthly-cycle listings), with additional nearby months also listed; active contracts extend approximately 1-2 years forward. |
| Settlement type | Physical delivery of platinum (99.95% minimum purity, NYMEX-approved brands/refiners) at NYMEX-approved depositories in the New York area. As with the COMEX precious metals, the substantial majority of contracts are cash-settled or rolled rather than taken to physical delivery. |
| Trading hours | CME Globex: Sunday-Friday 18:00-17:00 ET (23-hour session); liquid hours approximately 08:20-13:05 ET |
| Last trading day | Fourth-to-last business day of the delivery month |
| Contract value at $1,000/oz | $50,000 per contract |
| Related contract | Palladium futures (PA), also traded on NYMEX, 100 troy ounces -- the other major platinum group metal (PGM) with an actively traded futures market, and platinum's primary substitution partner described in Section 6. |
| LBMA/LPPM relationship | The London Platinum and Palladium Market (LPPM) -- historically the OTC reference market for PGM pricing, administering twice-daily platinum and palladium price fixes via the LBMA Platinum and Palladium Price, administered by ICE Benchmark Administration following the same general industry restructuring described for gold and silver benchmarks in the companion COMEX reference documents. |

### 1.1 Why the NYMEX/COMEX Distinction Matters in Practice

- While the practical trading and clearing experience is similar across CME Group's metals divisions, the NYMEX/COMEX division labelling persists in exchange rulebooks, regulatory reporting, and some data services -- meaning a trader requesting "COMEX platinum" data or documentation from certain sources may find the correct reference is technically NYMEX, a minor but occasionally consequential distinction when cross-referencing exchange rules or CFTC delivery and reporting categories.

## 2. Supply Concentration -- South Africa as the Dominant and Singular Source

Platinum's defining structural feature, more pronounced than for any other major metal covered in this document series, is the extreme geographic concentration of its mine supply in a single country and, within that country, a single geological formation -- giving platinum a country-specific political and operational risk exposure with few parallels among the metals covered elsewhere in this series.

| Country | Approx. Share of World Mine Production | Key Structural Facts | PL Curve Relevance |
|---|---|---|---|
| South Africa | ~70% of world production | The Bushveld Igneous Complex hosts the overwhelming majority of South African (and world) PGM reserves; production is deep-level, labour-intensive underground mining, with mines reaching among the greatest depths of any mining operation globally, raising both cost and operational complexity relative to most other metal mining | By far the dominant single-country supply variable; South African labour relations, electricity supply, and mine safety/operational disruptions are the primary supply-side risk factors for the entire global platinum market |
| Russia | ~10-15% of world production | Platinum is recovered substantially as a co-product of Russian nickel mining (notably in the Norilsk-Talnakh district), rather than from dedicated platinum-primary mines | A significant secondary source; sanctions-related market segmentation risk (described for Russian gold and silver in the companion COMEX reference documents) applies analogously to Russian platinum group metals |
| Zimbabwe | ~5-8% of world production | The Great Dyke geological formation hosts Zimbabwean PGM deposits, geologically related to but distinct from the Bushveld Complex | A smaller but non-trivial secondary African source, subject to its own country-specific political and operational risk factors |
| North America (Canada, US) | Remainder, single-digit percentage | Platinum recovered substantially as a co-product of nickel-copper mining (notably in Ontario's Sudbury Basin) rather than from dedicated platinum-primary operations | A minor supply contributor; the by-product nature of this supply source mirrors the by-product supply-elasticity limitation described for silver in the companion COMEX Silver reference document |

### 2.1 South African Labour Relations as a Recurring Supply Risk

- **The structural labour-intensity of deep-level mining**: South African platinum mining's deep-level, labour-intensive character (a consequence of the depth and geological character of the Bushveld Complex relative to more easily mechanised shallow or open-pit operations elsewhere in the mining industry) means labour costs and labour relations are a more central operational and cost variable for South African platinum mining than for many other major metal mining operations globally.
- **Documented historical strike disruptions**: South African platinum mining has, at various points, experienced significant labour action -- including strikes affecting major producers -- with the most severe documented episodes removing a material share of South African (and therefore global) production for extended periods, a recurring and well-documented supply risk factor specific to this market.

### 2.2 South African Electricity Supply (Load-Shedding) as a Mining Cost and Output Risk

- **Eskom and South African grid reliability**: South African state utility Eskom's well-documented electricity generation capacity constraints have, at various points over the past decade, resulted in scheduled and unscheduled power cuts ("load-shedding") affecting South African industrial users broadly, including platinum mining and processing operations, which require substantial and reliable electricity supply for underground ventilation, hoisting, and smelting/refining processes.
- **The mechanism's effect on platinum supply specifically**: load-shedding episodes have, at various points, forced South African platinum producers to reduce output or incur higher costs from backup power arrangements, an electricity-supply-driven supply risk factor layered on top of the labour relations risk described in Section 2.1 and largely specific to the South African operating environment among the major metals markets covered in this document series.

## 3. Demand Fundamentals -- Automotive Catalysis as the Dominant Application

Unlike gold, whose demand is overwhelmingly monetary/investment and jewellery-driven, platinum demand is dominated by a single industrial application -- automotive catalytic converters -- giving platinum a demand structure with more in common with the industrial-commodity framework described for copper and (in part) silver in the companion COMEX reference documents than with gold's monetary-asset framework.

| Application | Approx. Share of Total Platinum Demand | Key Driver | PL Curve Sensitivity |
| --- | --- | --- | --- |
| Automotive catalytic converters | ~35-40% | Diesel-engine vehicle catalytic converter loading specifically (platinum is the preferred catalyst metal for diesel applications, as distinct from gasoline-engine applications, which favour palladium -- see Section 6); global diesel vehicle production trends, particularly the European passenger diesel share decline described in the Gasoil reference document | The single most important demand category; European diesel passenger vehicle demand trends (declining structurally, per the Gasoil reference document) are a direct headwind for this specific platinum demand channel |
| Jewellery | ~20-25% | Significant in China and Japan specifically, where platinum jewellery has a more established consumer market than in some other major jewellery-consuming regions; price-sensitive, broadly similarly to gold jewellery demand but at platinum's own price level and with its own regional demand concentration | Chinese and Japanese consumer demand trends and relative platinum-gold pricing (since some substitution between platinum and gold jewellery exists at the margin) are relevant secondary demand signals |
| Industrial (chemical, glass, petroleum refining catalysts; electrical/electronics) | ~25-30% | Diverse industrial catalytic and electrical-contact applications, including use as a catalyst in chemical processing and petroleum refining, and in certain glass manufacturing processes | A diversified, generally more stable demand component distinct from the more cyclical automotive demand channel |
| Investment (bars, coins, ETFs) | ~5-10%, variable | Platinum-backed ETF holdings and physical bar/coin investment demand, smaller in scale than the equivalent gold investment market but responsive to similar macro/relative-value considerations, including the platinum-gold price ratio (Section 7) | A smaller but periodically significant swing factor in the overall demand balance |

| | | | |
|---|---|---|---|
| Hydrogen fuel cells (emerging) | Currently a small but structurally growing share | Platinum is a key catalyst material in proton-exchange membrane (PEM) hydrogen fuel cells, an application directly connected to the broader hydrogen economy and clean-energy transition theme | A long-horizon structural demand growth narrative analogous in framing (though much smaller in current absolute scale) to silver's solar photovoltaic demand growth story described in the companion COMEX Silver reference document |

## 3.1 The Diesel-Platinum Demand Link -- A Structural Headwind

- Because diesel-engine catalytic converters specifically favour platinum loading (relative to gasoline engines, which favour palladium, per Section 6), the well-documented structural decline in European passenger vehicle diesel share since the 2015 "Dieselgate" emissions scandal -- described in detail in the Gasoil reference document in this series -- represents a direct, multi-year structural headwind for platinum's largest single demand category, distinct from any cyclical automotive production variation and operating on the same long-run timeline as the broader European shift toward gasoline, hybrid, and electric vehicles.
- **Electric vehicle adoption as a longer-run consideration**: because battery electric vehicles require no catalytic converter at all, the broader global shift toward EV adoption represents a longer-run structural demand consideration for platinum's automotive catalyst demand overall (beyond the diesel-specific decline), a theme analysts weigh against the offsetting, though currently smaller-scale, hydrogen fuel cell demand growth narrative described in Section 3 (since hydrogen fuel cell vehicles, unlike battery electric vehicles, do use platinum as a catalyst material).

## 4. Exchange Inventories and the Physical Market

As with the COMEX precious metals, NYMEX warehouse stock data is a closely watched near-term physical market signal for platinum.

### 4.1 NYMEX Registered and Eligible Platinum Stocks

- NYMEX publishes daily registered (available for delivery) and eligible (meets delivery standards but not registered) platinum stock data, in the same general reporting format described for COMEX gold and silver stocks in the companion reference documents.

### 4.2 LPPM and London Vault Holdings

- As with gold and silver, London vault holdings data (administered under the LPPM framework referenced in Section 1) provides a broader view of global above-ground investment-grade platinum inventory beyond the NYMEX-specific registered/eligible figures.

## 5. The Forward Curve and Carry Structure

Platinum's forward curve is governed by a carry framework broadly similar to gold's -- financing cost less lease rate -- though the relative scarcity of platinum lease/borrow market liquidity compared to gold and silver, combined with platinum's industrial-demand exposure, can produce greater curve dislocations than are typical of the deeper, more liquid gold lease market.

### 5.1 Carry and the South African Supply Risk Premium

- As with silver's by-product mine supply inelasticity described in the companion COMEX Silver reference document, platinum's extreme single-country supply concentration (Section 2) means the market periodically prices in a supply-disruption risk premium during episodes of South African labour unrest or electricity supply disruption (Section 2.1-2.2), which can produce backwardation or curve flattening distinct from a pure interest-rate-driven carry calculation -- a phenomenon conceptually similar to the by-product supply-inelasticity-driven backwardation described for silver, here driven by single-country political/operational risk rather than by-product economics specifically.

## 6. Platinum-Palladium Substitution -- The Defining Cross-Metal Relationship

Because platinum and palladium are both viable catalyst metals for automotive catalytic converters, with the specific choice between them depending partly on engine type and partly on relative price, the platinum-palladium price relationship is the single most important cross-market relationship for understanding platinum demand.

### 6.1 Why Diesel Favours Platinum and Gasoline Favours Palladium

- **Catalytic chemistry differences**: diesel exhaust and gasoline exhaust have different chemical compositions and operating temperature profiles, and platinum has historically been the more effective and widely used catalyst for diesel oxidation catalysts specifically, while palladium has been preferred for gasoline-engine three-way catalysts -- an engineering/chemistry-driven specialisation rather than a pure cost-driven choice, though relative cost does influence the specific loading ratios used within each engine-type application.
- **The substitution that does occur**: within the bounds set by the chemistry described above, auto manufacturers and catalyst formulators have, at various points, adjusted the specific platinum-palladium-rhodium loading mix within a given catalyst formulation in response to relative metal prices -- meaning some substitution flexibility exists at the margin even though platinum and palladium are not fully interchangeable across the diesel/gasoline divide in the way, for example, Arabica and Robusta coffee are substitutable in blending applications (described in the Robusta Coffee reference document).

### 6.2 The Platinum-Palladium Price Ratio

- The platinum-palladium ratio (analogous in function to the gold-silver ratio described in the companion COMEX Silver reference document) is tracked as a relative-value signal, reflecting the relative tightness of the two metals' specific supply and demand balances -- platinum's heavier exposure to the diesel-demand decline (Section 3.1) and South African supply risk (Section 2) versus palladium's heavier exposure to gasoline-engine demand and Russian/South African co-product supply dynamics.
- **The historical price relationship shift**: platinum traded at a premium to palladium for most of their shared trading history, reflecting platinum's greater rarity and broader industrial/jewellery demand base; this relationship inverted for a sustained period in recent years as palladium demand (tied to gasoline-engine vehicle production, which grew as a share of global production even as diesel demand specifically declined) outpaced platinum demand -- a documented structural shift in the relative-value relationship that illustrates how the underlying demand-application differences described in Section 6.1 can produce a sustained reversal in a historically stable price relationship.

## 7. Cross-Market Relationships

### 7.1 Platinum and Gold

- The platinum-gold ratio is tracked as a secondary relative-value signal, given platinum's partial jewellery-demand overlap with gold (Section 3) and its periodic role as an alternative precious metals investment vehicle, though the relationship is generally regarded as looser and less reliable than the gold-silver ratio described in the companion COMEX Silver reference document, given platinum's much heavier industrial-demand weighting relative to gold's overwhelmingly monetary-asset profile.

### 7.2 The Monetary/Real-Rate Framework

- To the extent platinum is held as an investment asset (Section 3, investment demand category), it is subject to a modest version of the real interest rate and USD framework described in full in the companion COMEX Gold reference document, though this framework is considerably less dominant for platinum than for gold given platinum's much larger industrial-demand weighting -- meaning platinum should be analysed predominantly through an industrial-commodity lens (automotive demand, South African supply risk) with the monetary/real-rate framework as a secondary consideration, broadly the inverse weighting of the framework described for gold itself.

### 7.3 South African Rand (ZAR/USD)

- As with other major producer currencies described throughout this document series, ZAR/USD movement affects South African platinum mining cost economics: ZAR depreciation reduces South African miners' USD-equivalent operating costs at a given rand-denominated cost structure, improving margins and reducing the incentive to curtail production even at a lower USD platinum price, while ZAR appreciation has the opposite effect -- a mechanism operating in the cost/margin direction opposite to the revenue-side currency mechanisms described for agricultural exporters (BRL, VND, and others) elsewhere in this series, since platinum mining cost structures are domestic-currency-denominated while revenue is USD-denominated.

### 7.4 Speculative Positioning

- **CFTC COT for NYMEX Platinum**: weekly managed money net positioning is tracked, consistent with the positioning framework described throughout this document series. Platinum's smaller market size relative to gold means positioning extremes can, at times, be associated with sharper price moves than equivalent positioning levels in the much larger and more liquid gold market.

## 8. Documented Historical Events

- **South African platinum mining strikes (various episodes)**: documented labour action affecting major South African platinum producers has, at various points, removed a material share of South African production for extended periods, producing significant price reactions given South Africa's dominant share of global supply (Section 2).
- **Eskom load-shedding episodes (various years)**: documented periods of South African electricity supply constraint have affected platinum mining and processing output, as described in Section 2.2.
- **The platinum-palladium price relationship inversion**: as described in Section 6.2, the sustained shift from platinum's historical premium over palladium to a period of palladium trading at a premium to platinum is among the most significant documented structural developments in the PGM market over the past decade, directly tied to the diverging diesel-versus-gasoline demand trajectories described in Section 3.1 and Section 6.1.

## 9. Key Data Sources and Release Schedule

| Report / Data | Publisher | Frequency | Release Timing | Primary PL Curve Impact |
|---|---|---|---|---|
| South African platinum production/labour relations updates | Major producers (company disclosures); South African Chamber of Mines / Minerals Council | As reported/scheduled | Variable | All contracts -- the dominant single supply-side variable given South Africa's production share |
| Eskom load-shedding stage announcements | Eskom (South Africa) | As scheduled/announced | Variable, often daily during constrained periods | Front-to-mid contracts -- mining output risk during constrained electricity periods |
| World Platinum Investment Council (WPIC) quarterly market reports | WPIC | Quarterly | Variable | All contracts -- comprehensive global supply/demand balance, the primary structural reference |
| Global light vehicle production and diesel/gasoline/EV mix data | Industry associations; company disclosures (e.g., ACEA for European data, per the Gasoil reference document) | Monthly/periodic | Variable | Mid-to-back contracts -- structural automotive catalyst demand trajectory |
| NYMEX registered/eligible platinum stocks | CME Group | Daily | After close | Front contract -- delivery and physical tightness signal |
| LBMA/LPPM Platinum Price (AM/PM fix) | IBA via LBMA/LPPM | Twice daily | ~09:45 / 14:00 London | Spot reference -- physical market benchmark |
| ZAR/USD exchange rate | Reuters / Bloomberg / SARB | Daily/Continuous | Market hours | Cost-side input for South African mining economics (Section 7.3) |
| US Dollar Index (DXY) and 10-year TIPS yield | Bloomberg / Reuters / Fed | Continuous | Market hours | Secondary, monetary-framework input (Section 7.2) |
| CFTC Commitments of Traders (Platinum) | CFTC | Weekly | Fri ~15:30 ET (3-day lag) | Positioning / crowding -- given platinum's smaller market size, extremes can be more impactful than in gold |

NYMEX Platinum is, among the precious metals covered in this document series, the one most clearly governed by an industrial-commodity analytical framework rather than a monetary-asset one: its demand is dominated by automotive catalytic converter applications (with a documented structural headwind from the European diesel decline), and its supply is concentrated in a single country -- South Africa -- to a degree with few parallels among the other metals in this series, with deep-level mining labour relations and electricity supply reliability as recurring, country-specific operational risk factors. The platinum-palladium relationship, governed by the diesel/gasoline catalyst chemistry split, is the defining cross-market relationship, and its historic inversion over the past decade is among the most significant documented structural shifts in the broader PGM complex.

***For spread traders, the most reliable framework treats platinum as primarily an industrial-demand and single-country-supply-risk market, with the monetary/real-rate***

*framework relevant only at the margin (the inverse weighting from gold). South African labour and Eskom electricity developments, global diesel/gasoline/EV vehicle mix trends, and the platinum-palladium ratio are the three variables most worth tracking continuously, while the NYMEX/COMEX division distinction, though largely a bookkeeping matter in practice, is worth getting right when cross-referencing exchange documentation or regulatory data.*